EXP 4: To write C program to implement FCFS CPU scheduling algorithm.

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ vim fcfs_1290.c
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ gcc fcfs_1290.c
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ ./a.out
Enter number of processes: 3
Enter Arrival Time and Burst Time for P1: 0
3
Enter Arrival Time and Burst Time for P2: 2
4
Enter Arrival Time and Burst Time for P3: 4
6

P       AT      BT      CT      WT      TAT
P1      0       3       3       0       3
P2      2       4       7       1       5
P3      4       6       13      3       9

Average Waiting Time = 1.33
Average Turnaround Time = 5.67
Average Completion Time = 7.67
Throughput = 0.23 processes/unit time


Gantt Chart:

 ------------------------
|  P1   |  P2   |  P3   |
 ------------------------
0       3       7       13lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$
```

lab-03-17@lab-03-17-OptiPlex-3280-AIO: ~

```c
#include <stdio.h>

int main() {
    int n, i, j;
    int at[10], bt[10], ct[10], wt[10], tat[10], p[10];
    int time = 0, temp;
    float avg_wt = 0, avg_tat = 0, avg_ct = 0, throughput;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        p[i] = i + 1;
        printf("Enter Arrival Time and Burst Time for P%d: ", p[i]);
        scanf("%d %d", &at[i], &bt[i]);
    }

    // Sort by Arrival Time (FCFS)
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (at[i] > at[j]) {
                temp = at[i]; at[i] = at[j]; at[j] = temp;
                temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                temp = p[i];  p[i]  = p[j];  p[j]  = temp;
            }
        }
    }

    // FCFS calculation
    for (i = 0; i < n; i++) {
        if (time < at[i])
            time = at[i];

        time += bt[i];
        ct[i] = time;
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        // Add to totals
        avg_wt += wt[i];
        avg_tat += tat[i];
        avg_ct += ct[i];
    }
    // Calculate averages
    avg_wt /= n;
    avg_tat /= n;
```

Screenshot captured
You can paste the image from the clipbord.

```c
    // Calculate averages
    avg_wt /= n;
    avg_tat /= n;
    avg_ct /= n;

    // Throughput = number of processes / total time taken
    throughput = (float)n / (ct[n-1] - at[0]);
    printf("\nP\tAT\tBT\tCT\tWT\tTAT\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               p[i], at[i], bt[i], ct[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f", avg_tat);
    printf("\nAverage Completion Time = %.2f", avg_ct);
    printf("\nThroughput = %.2f processes/unit time\n", throughput);

        // Gantt Chart
    printf("\n\nGantt Chart:\n\n");

    // Print top bar
    printf(" ");
    for (i = 0; i < n; i++) {
        printf("--------");
    }
    printf("\n|");

    // Print process names
    for (i = 0; i < n; i++) {
        printf("  P%d   |", p[i]);
    }
    printf("\n ");

    // Print bottom bar
    for (i = 0; i < n; i++) {
        printf("--------");
    }
    // Print time line
    printf("\n%d", at[0]);
    for (i = 0; i < n; i++) {
        printf("      %d", ct[i]);
    }
    return 0;
}
```